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How disruptive is frugal?



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ABSTRACT

Global warming and the recent financial crises have prompted a rethink among businesses on the production of cost-effective products and services. *Frugal-innovations*, with their genesis mainly in India, are accordingly becoming popular in emerging and developed economies due to their lower costs and no frills structure. By studying a variety of frugal-innovations, this effort brings out the features of these innovations and also examines their disruptiveness vis-à-vis other innovations. Furthermore, this paper has formalized the *frugal-innovation* process based on examples covered in this effort. Such a formalization will aid in the metamorphosis of an ordinary innovation into the frugal type.

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1. Introduction

In recent years, there has been a growing awareness among companies of innovating with limited resources. Industry bellwethers, such as GE and Tata Sons, have successfully tapped into *frugal-innovations* to satisfy the needs of consumers at the base of the pyramid encompassing the low-income group of people. The Economist [28] has defined a frugal-innovation to be cheap, tough, easy to use and developed with minimal amounts of raw materials. Even some companies in Estonia, member of the European Union, have tapped into “minimalist” products to tide over the recent recession and, which they now successfully export [29]. According to Srinivas & Sutz [25], a scarcity-induced innovation is realized under scarcity in any one or combination of the following inputs: knowledge, institutions and socio-economics. Henceforth, this paper will refer to a scarcity-induced-, minimalist- or reverse-innovation as a frugal-innovation.

The use of frugality in emerging markets was first reported under “frugal-engineering” by Carlos Ghosn, CEO of Renault [39]. Since then *frugal-innovations* possessing a no frills structure have been developed for the thrifty consumer under constraints of developing countries. These innovations are realized under conditions that are not conducive for innovating in the rich world. The genesis of *frugal-innovations* is mainly ascribed to India and, China to a lesser extent, where improvisations are widely sought after for living under frugal conditions [28,30]. A quintessential example of *frugal-innovation* is *Jugaad*, a concept that has been adopted in management philosophy [13,23]. The *Jugaad* signifies a makeshift cart assembled in north India, under scarce resources, for handling routine chores. The growing significance of *frugal-innovation* is also attested to by its inclusion in the top ten management ideas of the millennium [11], with Immelt et al. [9] forecasting their export from China and India to the rest of the world.

The dawning of the 21st century has been accompanied by globalization along with natural and man-made-crises. First, independent institutions such as the Intergovernmental Panel on Climate Change (IPCC) have identified

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climate change as a serious threat to our way of life [7,8,15], which is supplemented by other threats including losses-in-bio-diversity and scarcer resources [24]. At the very least, these natural crises beg for a spartan lifestyle to preserve our complex ecosystems for the betterment of current and future generations. Man-made threats such as the dot-com bubble [14] and the recent financial crises [37] in particular, are also prodding the global populace toward thrift. Second, globalization is expanding the ranks of the working class in emerging economies, whose desire for higher standards of living [2,38] is putting a strain on our resources. Moreover, perturbations due to the above mentioned threats will get aggravated due to the forecasted explosion in world population [36].

The prospect of the global population turning abstemious, against the above backdrop, is dim due to the aspirations of an expanding middle class in emerging markets and also rich nations shrugging to opt for simpler lifestyles. An appealing solution to this dilemma is the use of *frugal-innovations* with cutting edge technology to create low-cost and no frills products. The adoption of frugality entails design principles that advocate minimal use of resources for realizing efficient functioning of products. Other than minimal amounts of raw materials, the economizing of resources would also involve the reuse of components and simpler designs that result in products without extra accessories. The no frills nature of frugal products makes them “lean” when compared to their “ordinary” counterparts, which are bulky due to consumption of relatively large amounts of resources. Consequently, aiming for a no frills product helps maintain lower costs with a positive impact on sustainability because of lesser resource consumption.

The importance of low price and sustainability is borne out by current trends. On the one hand, natural threats are empowering governments world over to enact stringent regulations on emissions and energy efficiency. This is exemplified by the adoption of carbon trading schemes and/or energy efficient technologies and/or emissions limits by various economies of the developed world and the BRICs (Brazil, Russia, India and China) [1,16,17]. On the other hand, cost-conscious consumers in emerging markets are demanding quality products and services at a discount for better lifestyles. Hence, companies will have to adopt the principles of *frugal-innovations* to achieve profitability against both cost-conscious consumers and regulations for sustainable-development.

By economizing the usage of resource-use and aiming for simple products and services, albeit sometimes using cutting edge technology, *frugal-innovations* have a lot to offer for sustainable-development. As a result, rich countries, such as the U.S., whose technological growth in recent years has flattened [4] should join the frugal bandwagon to give a boost to technological growth that will impact both economic growth and employment. Even export powerhouses like Germany with its legendary *Mittelstand* will have to embrace these innovations to cater to cost- and environment-conscious customers of the future.

However, systematic studies of *frugal-innovations* are scarce despite their significance against the current

backdrop. The effort by Prahalad and Mashelkar [22] describes *frugal-innovations*, under the banner of *Gandhian innovation*, but does not address the need to optimize ordinary innovations into the frugal type. This paper is an attempt in that direction, which lists the important features of *frugal-innovations* and also examines their disruptive potential. This effort also proposes a methodology for either creating *frugal-innovations* from scratch or transforming an arbitrary innovation into the frugal type.

2. Discussion

Tables 1–3 list some of the popular and not-so-popular *frugal-innovations* developed to date the world over. The terminology used in these tables furnishes as complete a set of features as possible, with simple technology comprising any or combinations of simple design, low part count, ease of assembly and other features. In contrast, sophisticated technology entails the use of cutting edge technologies to make frugal products. The term independent business team refers to the creation of an independent organization by either a company or an entrepreneur to engender a structure and culture conducive to frugal-innovations. Lastly, *sustainable design* refers to the reuse of existing components.

2.1. Features of frugal-innovations

The frugal products and services listed in Tables 1–3 are priced lower than their ordinary counterparts, which were designed by following standard design procedures. Table A-1 presents the cost and features of some ordinary products and services to aid in this comparison. The bamboo microscope, listed in Table 2, is another classic example of *frugal-innovation*, which at \$4 is cheap enough for use as a teaching aid and also one instance of basic research in India [21]. The theme of lower costs runs through the entire gamut of products and services discussed in this effort.

Besides the Tata Nano® and GE's Mac 400® and portable ultrasound, well known frugal products in engineering, the foray of *frugal-innovations* into the hi-tech sector is noteworthy. In particular, the electronics for the robotic hand at Sandia Laboratories were built from components found in cell phones [18] and the radio telescopes for the much vaunted *Square Kilometer Array* (SKA) project are satellite dishes retrofitted for use in astronomy [19]. This eye for frugality has resulted in significant cost savings as is evident from entries 9 and 10 in Tables 2 and A1, respectively. The frugal speech synthesizer, Avaz®, employs cutting edge technology for supporting multiple languages as opposed to the single language limitation of its pricier counterpart from the U.S. [26]. Even the U.S. Navy has developed low-priced Spike missiles, entry 11 in Tables 2 and A1, that are at par in performance with their premium counterpart, i.e., Javelin, in destroying targets such as armed trucks [34].

Despite their frugal nature, many of the products reported in this paper are also good in their quality.

Table 1
Popular examples of frugal-innovation.

No	Company/ organization	Country	Name of product	Type of product	Frugal features	Reference
1	Tata Motors	India	Tata Nano®	Car	a. Lower cost b. Sophisticated technology c. Ease of use d. Independent business team	[28]
2	Tata Chemicals Limited	India	Tata Swach®	Water Purifier	a. Lower cost b. Simple technology c. Ease of use d. Portable	[28]; http://www.tatachemicals.com/products/tata_swach.htm
3	General Electric	U.S. (Lab in India)	Mac 400® and Mac 800®	Handheld Electrocardiogram	a. Lower cost b. Sophisticated technology c. Portable d. Ease of use e. Robust f. Independent business team	[28]
4	General electric	U.S. (Lab in China)	–	Portable Ultrasound	a. Lower cost b. Sophisticated technology c. Portable d. Ease of use e. Robust f. Independent business team	[9]
5	General Electric	U.S. (Lab in India)	Well Baby Bassinet® and Infact Warmer Systems® (IWS)	Bassinet	a. Lower cost b. Sophisticated technology c. Portable d. Ease of use e. Independent business team	[9]; https://www2.gehealthcare.com/portal/site/usen
6	Nokia	Finland	Nokia 1200®	Mobile Handset	a. Lower cost b. Sophisticated technology c. Robust d. Ease of use	[28]; http://www.nokia.co.in/find-products/products/nokia-1200/technical-specifications
7	Godrej and Boyce Manufacturing	India	ChotuKool®	Refrigerator	a. Lower cost b. Sophisticated technology c. Portable d. Ease of use	[28]; http://www.chotukool.in/
8	Bharathi Airtel	India	–	Mobile Services Operator	a. Lower cost b. Large-scale outsourcing c. Independent business team	[28]

The Bamboo Bicycle's ride comfort has been reported to exceed that of its metallic counterpart [12]. The good quality of the portable electrocardiogram, the portable ultrasound and the robotic hand are reflected in their good performance. In fact, GE's portable products are widely used in developed and developing countries [9] and Sandia's robotic hand has been reported to perform intricate tasks thus making it suitable for critical applications [18]. The malfunctioning observed in a few of the frugal products, i.e., the Tata Nano® [33,35] and the telescopes in the SKA project, is no cause for concern. These malfunctions were limited to a few cars in the case

of the Tata Nano and just one radio telescope in the SKA project. The Tata Nano 2012®, a second generation frugal product, is a reliable and improved version of the original concept and the breakdown of the telescope boils down to the non availability of spares. In general, frugal products and services will improve with successive generations and, the few concerns in terms of design and spares seen in some of the current versions will be ironed out in the future.

Frugality has also influenced the fight against diseases, with researchers at Stanford University in the U.S. having developed a novel compact probe for detecting

Table 2Not-so-popular examples of *frugal-innovation*.

No	Company/ Organization	Country	Name of Product	Type of Product	Frugal Features	References
9	Sandia National Laboratories	U.S.	–	Robotic Hand	a. Lower cost b. Robust c. Sustainable design d. Sophisticated technology e. Ease of use	[18]
10	South Africa	South Africa	Square Kilometer Array	Radio Telescope	a. Lower cost b. Robust c. Sustainable design d. Sophisticated technology	[19]
11	Naval Air Weapons Station	U.S.	Spike	Missile	a. Lower cost b. Robust c. Sophisticated technology	[34]
12	Lunar Design	U.S.	–	Bone Drill	d. Ease of use a. Lower cost b. Robust c. Simple technology d. Ease of use e. Start up	http://www.fastcodesign.com/1664799/stanford-finds-hints-for-the-future-of-us-medical-design-in-india ; http://www.lunar.com/a-new-model-for-medical-device-development/
13	Columbia University (Earth Institute)	U.S.	–	Bamboo Bike	a. Lower cost b. Robust c. Simple technology d. Ease of use e. Start up	[12]
14	Tata Consultancy Services (TCS)	India	Home Infotainment Platform (HIP) [®]	An information and communication platform providing internet gateway on a television set.	a. Lower cost b. Tough or robust c. Sophisticated technology d. Ease of use	[28]; http://www.tcs.com/resources/brochures/Pages/Home-Infotainment-Platform.aspx
15	MittiCool	India	MittiCool Fridge [®]	Refrigerator	a. Lower price b. Zero maintenance cost c. Simple technology with zero energy consumption (earthen pot principle) d. Start up	[5]; http://www.mitticool.in
16	Jodo Gyan	India	–	Bamboo Microscope	a. Lower price b. Simple technology c. Start up	[21]
17	RanBaxy Laboratories Limited	India	–	Generic drugs	a. Lower price b. Sophisticated technology	[27]
18	Narayana Hrudayalaya Hospital	India	Health City [®]	Healthcare	a. Economies of scale (large volumes of surgeries) b. Low price surgery c. Specialized or sophisticated surgery	[10,28]
19	Wockhardt Hospitals	India	–	Beating Heart Surgery	a. Economies of scale (large volumes of surgeries) b. Low price surgery c. Specialized or sophisticated surgery	[27]
20	Aravind Eye care System	India	–	Eye care	a. Low price surgery b. Economies of scale (large volumes of surgeries) c. Sophisticated technology	[27]
21	Columbia Asia	U.S.	–	Health care	a. Low price services b. Sophisticated technology using healthcare information technology (HIT)	[27]
22	First Energy	India	OORJA [®]	Stove and Fuel	a. Lower cost b. Lesser energy consumption and emissions c. Simple technology d. Start up	[28]; http://www.pciaonline.org/first-energy-private-limited

Table 2 (continued)

No	Company/ Organization	Country	Name of Product	Type of Product	Frugal Features	References
23	A Little World	India	–	Rural banking through mobile phony	a. Lower cost b. Sophisticated technology c. Portable d. Ease of use e. Robust f. Start up	[28]; http://www.alittleworld.com/
24	Invention Labs	India	Avaz®	Speech synthesizer	a. Lower cost b. Sophisticated technology c. Supports multiple languages d. Portable e. Ease of use f. Robust g. Start up	[26]
25	Indian Government	India	Aakash UbiSlate 7®	Touchscreen Tablet Computer	a. Lower cost b. Sophisticated technology c. Portable d. Ease of use e. Robust	http://www.ubislate.com/ ; http://www.bbc.co.uk/news/world-south-asia-15180831
26	Mindray	China	–	Healthcare products such as monitoring and life support, diagnostics and imaging.	a. Lower cost b. Sophisticated technology c. Portable d. Ease of use e. Robust f. Start up developed into a full-fledged company	[28,31]
27	BYD	China	–	Lithium-ion Battery	a. Lower cost b. Sophisticated technology c. Start up developed into a full-fledged company	[28]
28	Dachangjiang	China	–	Motorcycle	Lower cost through flexible supply-chain network	[28]

tuberculosis (TB) [20]. This highly efficient probe comprising a mobile phone camera and light box detects smaller numbers of TB bacteria in a small sample when compared to existing testing procedures that are not cost effective.

In addition to lower costs, sustainability and ease of operation, a bulk of the products in these tables has been realized from cutting edge technology. Accordingly, the product-based features of a *frugal-innovation* based on the discussion above are: low price, compact design with a no frills structure, use of limited raw materials or reuse of existing components, ease of use and use of cutting edge technology, where possible, to achieve lower costs. An independent organizational structure with a culture promoting frugality is also evident in the creation of these innovations.

2.2. Are frugal-innovations disruptive?

Ordinarily businesses improve their profitability by undertaking various cost cutting measures. Traditionally these post design measures are directed at minimizing the cost of a given product by streamlining the various processes that go into its realization. In contrast, cost cutting is built into the initial design process of a *frugal-innovation*, thereby producing products that are cheaper than the cheapest ordinary products. Moreover, the good quality of a majority of the frugal products and, the quality improvement seen in some second generation products, puts *frugal-innovations* in a formidable position to disrupt their incumbents. This subsection examines the disruptiveness by applying the principles propounded by Christensen [3] on disruptive innovations.

Table 3

Frugal-innovations in the making.

No	Company/organization	Country	Name of product	Type of product	Frugal features	References
29	Tuck School of Business, Dartmouth College	U.S.	300\$ House	Housing for the poor	a. Lower cost b. Simple design c. Minimum features for living with dignity.	[6,32]
30	Stanford University	U.S.	–	Probe for Detecting Tuberculosis	a. Lower cost b. Simple design c. Portable	[20]

According to Christensen [3], a disruptive innovation is cheaper, smaller, simpler and convenient to use. Aside from these product-based features, Christensen [3] also lists characteristics pertaining to the relevant markets, organizational structure and resource allocation. In addition to lower costs, most of the products listed in Tables 1–3 are compact in their design and/or smaller and have all been noted for ease of use. However, products such as the Mac 400® and Mac 800® extend the concept of Christensen's disruption by being sophisticated in their technology.

A crucial distinction between Christensen's disruptive innovation and the frugal type involves the evolution of performance attributes. The former starts with a product attribute such as compactness or ease of use and ends with lowering of price through product commoditization. The latter however, is based on low prices and would evolve in the future by optimizing other attributes. The emphasis on lower cost restricts the designer from using resources in excess of that required for a *frugal-innovation*. Each subsequent optimization of other features, such as quality, streamlines the frugal-design process. The end result is a product comprising bare minimum of components, which functions at par with, or even better than, the traditional counterpart. In addition, the precedence of lower cost gives a head start and, more time, to a *frugal-innovation* to evolve into a full-fledged lean product with good functionality. In contrast, a traditional product might not mature into a leaner version.

The rubric of lower cost running through the examples of Tables 1–3 make it a crucial factor aiding disruption of ordinary innovations in the future. With the continuing growth of emerging economies, the ranks of their frugal consumers will continue to swell. Therefore, in light of weaker demand in most western economies, companies worldwide will seriously consider these disruptions by *frugal-innovations* and vie with each other to capture larger shares in emerging markets. In other words, a variation of Christensen [3] idea can be observed where consumers influence mainstream companies to adopt frugality into their value networks.

Besides lower price, the no frills aspect of frugal-innovations lends itself to the needs of sustainability for tackling the planetary crises of our time. The *factor-of-safety* traditionally applied in designs typically leads to greater consumption of resources. However, the *factor-of-frugality* inherent in *frugal-innovations* would result in significant savings of resources when manufacturing many of these innovations for mass consumption. The economic utilization of resources has a positive impact on society and would aid in preserving resources for future generations. Another positive impact on sustainability would be the foreseeable improvement in the quality and, life, of frugal products thus optimizing consumption. Overall, the lower cost together with no frills design and good quality will create hoards of frugal products and services in the future that are beneficial from the perspective of both cost and sustainability. Frugal products will raise the living standards of society at large by making their good quality performance affordable at lower costs. The powerful combination of

low-cost, quality and simplicity makes mass consumption of frugal products a sustainable solution for a better tomorrow. Therefore, *frugal-innovations* are a significant disruptive force.

3. Methodology for undertaking frugal-innovations

The *frugal-innovation* process, as carried out in emerging markets, typically involves tenacious entrepreneurs working against a backdrop of scarce resources such as knowledge, raw materials, personnel and a rule of law conducive to innovations, at least in the past. The process description that follows presents a suitable methodology for formalizing the *frugal-innovation* process that can aid in developing these products in various sectors including engineering and science. While there is interaction between the individual steps, the discretization clarifies understanding of the methodology and also facilitates its systematic application to a problem. It should be noted that this section uses design and innovation interchangeably since a frugal variant of an existing product qualifies for an innovation.

3.1. Modeling of functionality

The first step of the proposed methodology is the detailed design process using *classical* analyses. A classical analysis refers to the use of traditional scientific, engineering and other relevant techniques generally used for designing products. Besides understanding the viability of making a product frugal, such a study would serve as a reference for optimizing the basic design for frugality. Even products in critical sectors, where “diluting” a design might have disastrous consequences, should avail of *classical* analyses to determine the feasibility of undertaking the *frugal-innovation* process. Such a study should list all the parameters influencing the functionality of a given product along with their suitable values.

3.2. Frugal product feasibility

The second step of this methodology is examining the feasibility of creating a frugal version of a given product. In particular, results of the study undertaken in the previous step would be utilized for inferring the impact of frugality on the functionality of a given product. The functionality of some products in critical sectors, such as aerospace, defense and even healthcare, might not lend itself for alteration due to the superior and reliable performance expected in these sectors. However, apart from such cases, the majority of goods and services available for consumption are ready for frugality.

3.3. Optimization of basic design

The third step concerns the optimization of a given innovation against constraints of frugality. With reference to Fig. 1, a given innovation should be optimized for lower cost against a fixed constraint of functionality or performance, as determined in step 1 and, variable constraints

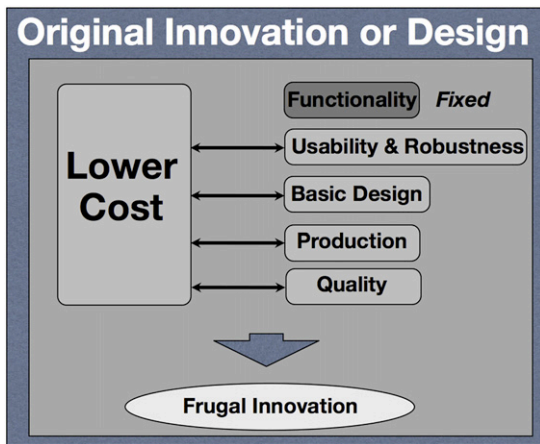


Fig. 1. Optimizing the basic design or innovation to frugal type. Note: Functionality (performance) is the fixed constraint while basic design, production, quality and usability and robustness are variable constraints.

such as basic design, production, quality, robustness and ease of use. The rationale being that optimization of the basic design for lower cost would result in a product synthesized through efficient utilization of resources, i.e., a *frugal-innovation*.

Each of the variable constraints could be controlled, in turn, through a wide range of factors. For instance, production encompasses manufacturing processes, process parameters, labor, supply-chain logistics and other relevant manufacturing variables that could be optimized for achieving lower costs. Furthermore, as depicted in Fig. 2, multiple iterations of variable constraints would be required in the general case for creating a frugal-innovation.

The objective could be extended to include sustainability to account for the product's impact on environment. Sustainability could also be built into the model as a fixed

constraint comprising sustainable materials, such as bamboo etc. and, *sustainable-design* principles. However, any redundancies arising from the implicit inclusion of this factor in the optimization process should be avoided.

The three-step methodology developed above could be “hardwired” into the innovator who will evaluate the constraints continuously during the innovation process. The rural *frugal-innovators* of India are the representatives of this approach. Alternatively, an innovative design or idea could be optimized, through this formalization, after its conception but well before production.

Yet another strategy for companies to engender frugal-innovations is maintenance of close proximity with the base of the pyramid. In other words, companies should direct a portion of their research and development budget to monitoring the lifestyles of both the low-income people and poor. These associations with the thrifty would spawn new products and services, improve existing products and services and, maybe, lead to newfangled concepts and models for frugality-based businesses.

4. Conclusions

Globalization together with an expanding population is straining earth's natural resources while also contributing to climate change. This paper has highlighted the significance of *frugal-innovations* against this backdrop. The development of no frills products and services has been showcased with thirty examples in a range of sectors including healthcare, banking, engineering and computer, to name a few. The strong tendency of *frugal-innovations* to disrupt incumbents was demonstrated through specific examples. A methodology was also presented for the systematic development of *frugal-innovations*. This methodology would aid companies in harnessing frugality and improving their profitability in a world conscious of both cost and sustainability.

Appendix

The features and prices of selected ordinary products are listed in this appendix for comparison with their frugal counterparts. These products are listed in the same order, i.e., same serial number, as in Tables 1–3 to facilitate comparison. All columns refer to the ordinary products except for “Frugal Cost”, which lists the price for the corresponding frugal entry in Tables 1–3. The costs of these ordinary versions are primarily based on existing literature where available and complemented by current prices, based on current exchange rates, procured from India's popular on-line shopping sites *flipkart* and *naaptol*.

The performance of frugal products, with reference to Table A1, is at par or better than their ordinary counterparts. The descriptions of some of these superior, i.e., better quality, frugal products are given in the section on discussion. Moreover, the column on features lists only the relevant characteristics that make these ordinary products distinct from the frugal type.

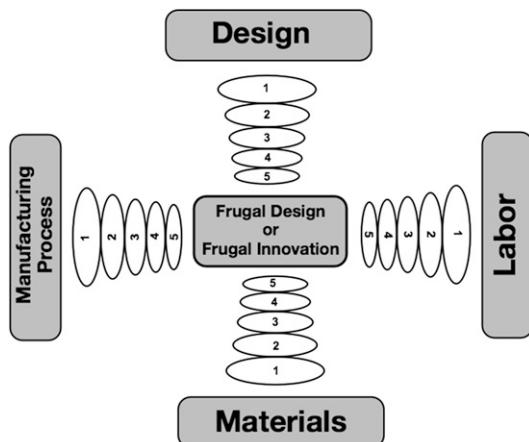


Fig. 2. Converging iterations of multiple variable constraints in the optimization process for creating a frugal product or *frugal-innovation*. Note: Each shrinking oval represents convergence towards optimal value for a given variable, with a total of 5 iterations shown here for clarity.

Table A1. Ordinary products and services

No	Product/service	Ordinary cost	Frugal cost	Ordinary features	References
1	Small Car (Model: <i>Figo</i> for the Indian market)	\$7370	\$2900	Higher cost	http://www.zigwheels.com/
2	Water purifier	\$270	\$23	a. Higher cost b. Intricate design	http://www.flipkart.com/
3	Electrocardiogram (<i>Marquette</i> series)	\$2000	\$800	a. Higher cost b. Intricate Design c. Bulky	[28]
4	Conventional Ultrasound	≥\$100,000	\$15,000	a. Higher cost b. Intricate Design c. Bulky	[9]
7	Refrigerator (lower capacity)	\$185	\$70	a. Higher cost b. Intricate Design c. Bulky	[28]; http://www.flipkart.com/
9	Robotic Hand	\$250,000	\$10,000	a. Higher cost b. Intricate design	[18]
10	Radio Telescope	>\$6 million	<\$2.5 million	Higher cost	[19]
11	Javelin Missile	\$147,000	\$5000	Higher cost	[34]
16	Microscope (<i>educational & some basic research</i>)	\$15 (<i>single lens</i>)	\$4	a. Higher cost b. Intricate Design	[21]; http://www.naaptol.com
18	Heart Surgery	\$20,000–\$1,00,000	\$2000	Higher cost	[10,28]
24	Speech Synthesizer	\$5000–\$10000	\$800	a. Higher cost b. Supports only one language	[26]
25	Tablet Computer (<i>iPad</i>)	\$570	\$80	Higher cost	http://www.ubislate.com/ , http://www.apple.com/in/ipad/
29	Affordable housing	\$3000 (<i>in the U.S.</i>)	\$300	Higher cost	[6,32]

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